

**GIS-BASED IDENTIFICATION OF WATER HARVESTING AND
GROUNDWATER RECHARGE SITES FOR SUSTAINABLE PADDY
CULTIVATION IN SAINJI WATERSHED, TEHRI GARHWAL,
USING GIS AND MACHINE LEARNING**

Submitted by

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Declaration

I hereby declare that the project report titled “GIS-Based Identification of Water Harvesting and Groundwater Recharge Sites for Sustainable Paddy Cultivation in Sainji Watershed, Tehri Garhwal, Using GIS and Machine Learning” is an original piece of work carried out by us.

The GIS analysis, machine learning workflow, and this written report represent our own independent effort, undertaken using freely available geospatial datasets and open-source software. Any material drawn from external sources has been duly acknowledged and cited in the References section.

This work has not been submitted, in part or in full, for the award of any other degree or diploma at this or any other institution.

Khushi jaiswal and Shagun chand

Acknowledgement

This project would not have been possible without the openness of the geospatial and scientific community that makes high quality data and tools freely available to independent researchers and students. I would like to acknowledge NASA and the Shuttle Radar Topography Mission (SRTM) team for the elevation data that underpins the terrain analysis in this study, the European Space Agency for Sentinel-2 imagery, and NASA POWER for the rainfall data used in the hydrological assessment.

I am grateful to the developers and maintainers of QGIS, Python, and the open-source scientific computing ecosystem (pandas, scikit-learn, matplotlib), whose tools made it possible to carry out a complete GIS-to-machine-learning workflow without any proprietary software.

Abstract

Groundwater is the lifeline of paddy cultivation in the Himalayan watershed of Sainji, Tehri Garhwal, where uneven terrain and a short, intense monsoon season make water availability the single biggest constraint on farming. This project sets out to identify where, within the watershed, rainwater is most likely to recharge groundwater, using a combination of Geographic Information System (GIS) analysis and machine learning, and relying entirely on free, open-access data and open-source software.

Digital Elevation Model (DEM) data from SRTM, Sentinel-2 optical imagery, and NASA POWER rainfall records were processed in QGIS to derive four thematic layers: slope, distance from the stream network, land use/land cover (LULC), and rainfall. These were reclassified into suitability scores and combined through a weighted overlay (slope 0.4, stream distance 0.3, LULC 0.2, rainfall 0.1) to produce a groundwater recharge suitability map for the watershed. Sample points were then generated across the study area, and the raster values at each point were exported as a training dataset for machine learning.

A Random Forest Regressor was trained in Python to predict recharge suitability from the Stream, Slope, and LULC layers. After cleaning (300 raw samples reduced to 253 complete records) and an 80:20 train-test split, the baseline model achieved an R^2 of 0.997, which improved marginally to 0.998 after hyperparameter tuning with GridSearchCV. Feature importance analysis identified LULC as the dominant predictor (60%), followed by slope (28%) and stream proximity (12%). Because the recharge scores used as the machine learning target were themselves derived from the same GIS layers used as predictors, this near-perfect accuracy reflects the model correctly re-deriving a known weighted formula rather than an independent validation against field-measured groundwater data—a limitation that is discussed openly in this report rather than glossed over.

The resulting suitability map highlights zones of high, moderate, and low recharge potential across the Sainji watershed, offering a practical, low-cost starting point for prioritising water harvesting structures such as farm ponds, infiltration trenches, and contour bunds. The complete workflow, code, and data are version-controlled and published on GitHub, making the analysis fully reproducible and adaptable to other small Himalayan watersheds facing similar water security challenges.

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Chapter 1: Introduction

1.1 Background

Across the Indian Himalaya, groundwater depletion is no longer a distant concern, it is something farmers in small watersheds like Sainji encounter every dry season. Paddy cultivation, which is the backbone of the local agricultural economy, depends on a steady supply of water at precisely the times when the region's rainfall is least reliable: the premonsoon and early monsoon windows when nurseries are raised and transplanting takes place. In flat alluvial plains, groundwater recharge can be estimated with relatively simple hydrogeological models. In steep, terraced, and geologically variable terrain such as Sainji, however, the picture is far more complex: recharge potential depends on the interplay of slope, soil, land cover, and proximity to drainage channels, all of which vary sharply over short distances.

In such terrain, the most direct and cheapest way of improving water security is by harvesting rainwater deliberately, that is, by capturing and storing monsoon runoff in farm ponds, infiltration trenches or contour bunds. But putting up these buildings in the wrong places is throwing away effort and money. What is needed is a way of identifying, before any earthmoving, the parts of the watershed that are naturally best suited to recharge. This is precisely the kind of spatial decision problem that Geographic Information Systems (GIS) were created to handle. In recent years, machine learning has emerged as a complementary tool that can validate and improve GIS-based suitability assessments by learning the relationships between terrain, land cover and recharge potential directly from the data.

1.2 Problem Statement

Sainji village, located in the Tehri Garhwal district of Uttarakhand, sits within a Himalayan watershed where sustainable paddy cultivation depends heavily on groundwater availability and effective rainwater harvesting. The uneven terrain and highly seasonal rainfall of the region make it difficult to identify suitable groundwater recharge zones through manual field observation alone; the process is slow, subjective, and hard to repeat consistently across a growing season, let alone across years. This

project addresses that gap by integrating GIS-based spatial analysis with a machine learning validation step, relying entirely on freely available geospatial datasets and open-source software so that the approach remains accessible and reproducible for other small watersheds facing similar constraints.

1.3 Objectives

- To delineate the boundary and drainage network of the Sainji watershed using SRTM elevation data.
- To generate thematic layers like slope, stream distance, land use/land cover, and rainfall relevant to groundwater recharge.
- To perform a weighted overlay analysis in QGIS and produce a groundwater recharge suitability map for the watershed.
- To apply a Random Forest machine learning model to validate and cross-check the relationship between the GIS-derived factors and recharge suitability.
- To develop a fully reproducible, open-source workflow, documented and version-controlled on GitHub, that can be adapted to other small Himalayan watersheds.

Chapter 2: Study Area

The study area is a small headwater sub-watershed surrounding Sainji village in the Tehri Garhwal district of Uttarakhand, India, centred at approximately 78.97°E longitude. As the study area map below shows, the watershed is a compact, steeply terraced catchment based on the scale bar in the mapped output, it spans roughly half a kilometre to a kilometre across, which is typical of a first order headwater catchment in this part of the Himalaya. (The exact area in hectares can be read directly from the GIS attribute table.)

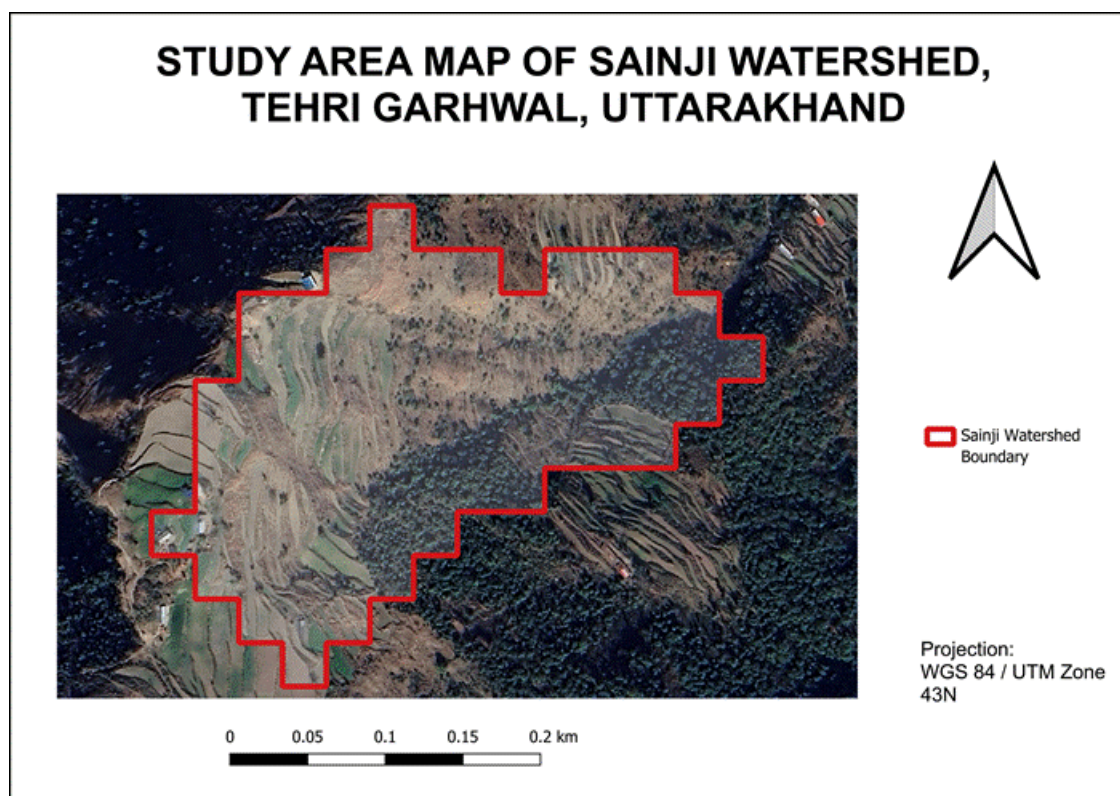


Figure 1: Study Area Map of Sainji Watershed, Tehri Garhwal, Uttarakhand (satellite base, watershed boundary in red)

2.1 Geographic Location and Terrain

The boundary of the watershed is marked on a satellite base image of a mixed landscape of terraced agricultural fields interspersed with patches of forest. The land use pattern is typical of subsistence hill agriculture in Garhwal where cultivation is concentrated on the gentler slopes, which are more accessible, and the steeper faces are covered with trees. This land cover mix has direct implications for groundwater

recharge, as terraced cropland, forest, and grassland impact infiltration in very different ways, a relationship explored further in Chapters 4 and 5.

2.2 Climate and Hydrology

Sainji shares a monsoon dominated rainfall regime with much of the mid-elevation Garhwal Himalaya . Most of the annual precipitation falls from June to September. Outside this window, water availability is very much a function of residual soil moisture and whatever groundwater has been recharged during the previous monsoon which is precisely why identifying and protecting high-recharge zones matters so much to sustain paddy cultivation through the drier parts of the year. The stream network of the watershed (see Figure 5.10, drainage and stream-distance analysis in Chapter 5) drains the terraced slopes to the lower reaches of the catchment and proximity to this stream network is one of four factors used in the recharge suitability model.

2.3 Agriculture and Land Use

Paddy remains the dominant crop across the lower and mid elevation terraces of the watershed, supplemented by other seasonal crops on the steeper or less irrigated plots. Because paddy is particularly water intensive relative to other hill crops, the sustainability of this cropping pattern is closely tied to the watershed's capacity to recharge groundwater during the monsoon and retain it through the growing season which is the central motivation for this study.

Chapter 3: Data Used

Every dataset used in this project is freely and publicly available, which was a deliberate choice. One of the goals of this project was to demonstrate that a credible, GIS-and-machine-learning-based recharge assessment does not require expensive commercial data or software.

The table below summarises the four primary datasets and their role in the analysis.

Dataset	Source	Resolution	Purpose in this Study
Digital Elevation Model (DEM)	SRTM (NASA/USGS)	30 m	Elevation, slope generation, watershed delineation, drainage/stream network extraction
Sentinel-2 Optical Imagery (Bands B02, B03, B04, B08)	European Space Agency (ESA)	10 m	Land Use / Land Cover (LULC) classification
Rainfall (Precipitation Sum Average)	NASA POWER	Point / gridded, station-derived	Rainfall thematic layer for the weighted overlay
Stream Network	Derived from DEM (hydrological analysis)	30 m (from DEM)	Stream distance layer; drainage pattern reference

3.1 Digital Elevation Model (DEM)

The 30 m resolution SRTM DEM is the backbone of the whole GIS workflow. The watershed boundary, slope map and stream network are all derived from this single dataset using standard hydrological processing in QGIS (fill sinks, flow direction, flow accumulation, and stream extraction, described in detail in Chapter 4). SRTM data, a global product not specifically surveyed for Sainji, has been commonly and successfully used in similar Himalayan watershed studies (Zimik et al., 2022), giving confidence that it is fit for purpose at this scale.

3.2 Sentinel-2 Imagery

Sentinel-2's 10 m resolution and four relevant bands (blue, green, red, and near-infrared) make it well suited to distinguishing forest, cropland, and grassland, these three LULC classes used in this study. The near-infrared band (B08) in particular is valuable for separating healthy vegetation from bare or cultivated land, since vegetation reflects strongly in that part of the spectrum.

3.3 Rainfall Data (NASA POWER)

NASA POWER provides long-term precipitation records derived from satellite and reanalysis data, which is especially useful in a data-sparse mountain region like Tehri Garhwal where ground-based rain gauge coverage is limited. The precipitation sum average was used to generate the rainfall thematic layer, which carries the smallest weight (0.1) in the overlay, reflecting the relatively lower spatial variability of rainfall compared to the sharply changing terrain factors within such a small watershed.

3.4 Stream Network

Rather than relying on an external hydrography dataset, the stream network was derived directly from the DEM using QGIS's hydrological tools. This keeps the entire workflow self-consistent; the same elevation surface that produces the slope map also produces the drainage network, so the two layers are guaranteed to be spatially coherent with one another.

Chapter 4: Methodology

The methodology followed in this project has two connected halves: a GIS workflow carried out in QGIS that produces a physically-grounded recharge suitability map, and a machine learning workflow carried out in Python that trains a Random Forest model on samples drawn from that same map. The overall flow of the study is summarised below.

4.1 Overall Workflow

- Data Collection : DEM (SRTM), Sentinel-2 imagery, NASA POWER rainfall data
- DEM Processing : fill sinks, flow direction, flow accumulation
- Watershed Delineation : boundary extraction from the processed DEM
- Slope Generation : slope raster derived from the DEM
- Stream Extraction : drainage network derived from flow accumulation
- LULC Processing : classification of Sentinel-2 imagery into forest, grassland, and cropland
- Stream Distance Analysis : Euclidean distance raster from the stream network
- Reclassification : each thematic layer scored on a common suitability scale
- Weighted Overlay : combination of the reclassified layers into a single suitability surface
- Recharge Suitability Mapping : continuous suitability index and classified suitability zones
- Sample Point Generation and Raster Sampling : export of a point-based training dataset
- Machine Learning Validation : Random Forest training, tuning, and evaluation in Python
- GitHub Repository Development version control and public documentation of the full workflow

4.2 DEM Processing and Watershed Delineation

The SRTM DEM was initially processed to fill artificial depressions (“fill sinks”) that would otherwise interrupt the simulated flow of water over the surface. The flow direction was then calculated for each cell according to the standard eight-direction (D8) method, and then flow accumulation was calculated, which is the amount of upstream

area that drains through each cell. Cells with flow accumulation above a threshold were extracted to represent the stream network. The watershed boundary was delineated by tracing the catchment that drains to the outlet point of interest near Sainji village. .

4.3 Factor Preparation

Four factors were prepared as inputs to the suitability model, each reflecting a different physical control on groundwater recharge:

- Slope: steeper slopes shed rainwater as runoff rather than allowing it to infiltrate, so gentler slopes score higher for recharge suitability.
- Stream Distance : areas closer to the drainage network tend to have shallower water tables and easier connectivity to subsurface flow, so proximity to streams scores higher.
- Land Use / Land Cover (LULC) : forest and grassland generally support higher infiltration than compacted or heavily cultivated cropland, though the relationship also depends on soil and tillage practices.
- Rainfall : areas receiving more rainfall naturally have more water available to recharge, all else being equal.

4.4 Reclassification and Scoring

Each of the four raster layers was reclassified onto a common suitability scale for meaningful combination. Slope, for example, was divided into classes from gentle (high suitability) to steep (low suitability). Stream distance was divided into proximity bands. LULC classes were ranked by their typical infiltration behavior. Rainfall was divided into bands that reflect the range observed across the watershed. The exact class boundaries and scoring tables used in QGIS's Reclassify tool can be included here from the project's QGIS model documentation for full transparency.

4.5 Weighted Overlay

The four reclassified layers were combined together using weighted linear addition, in which weights represent their level of significance regarding the role played by each of

the four factors in the process of groundwater recharge according to project assumptions:

$$\text{Recharge suitability} = 0.4 \times \text{Slope} + 0.3 \times \text{Stream distance} + 0.2 \times \text{LULC} + 0.1 \times \text{Rainfall}$$

The greatest weight (0.4) was assigned to the factor of slope because, in such rugged terrain, it is the most powerful factor determining whether rainwater will either infiltrate into the ground or run off the land surface. The next factor is stream distance, which is estimated as 0.3. The remaining factors LULC and rainfall receive relatively lower weights, of 0.2 and 0.1 respectively, because the spatial variation of rainfall in this small watershed is insignificant compared to other topographical factors. This weighing procedure is analogous to that used in other GIS-based recharge assessments in the Himalayas, such as that involving the application of the AHP method for assessing recharge criteria in the Sikkim Himalayas (Zimik et al., 2022).

4.6 Sample Generation and Export for Machine Learning

To bridge the GIS analysis and the machine learning stage, random sample points were generated across the watershed, and the underlying raster values like Slope, Stream distance, LULC, and the computed Recharge suitability score were extracted at each point using QGIS's point sampling tool. This point-based table was then exported as a CSV file, forming the training dataset used in Chapter 6.

Chapter 5: Results and Discussion

This chapter walks through the outputs of the GIS analysis, layer by layer, before bringing them together in the final recharge suitability and classification maps. Each

map was generated in QGIS from the SRTM DEM, Sentinel-2 imagery, and the derived stream network described in Chapters 3 and 4.

5.1 Elevation (DEM)

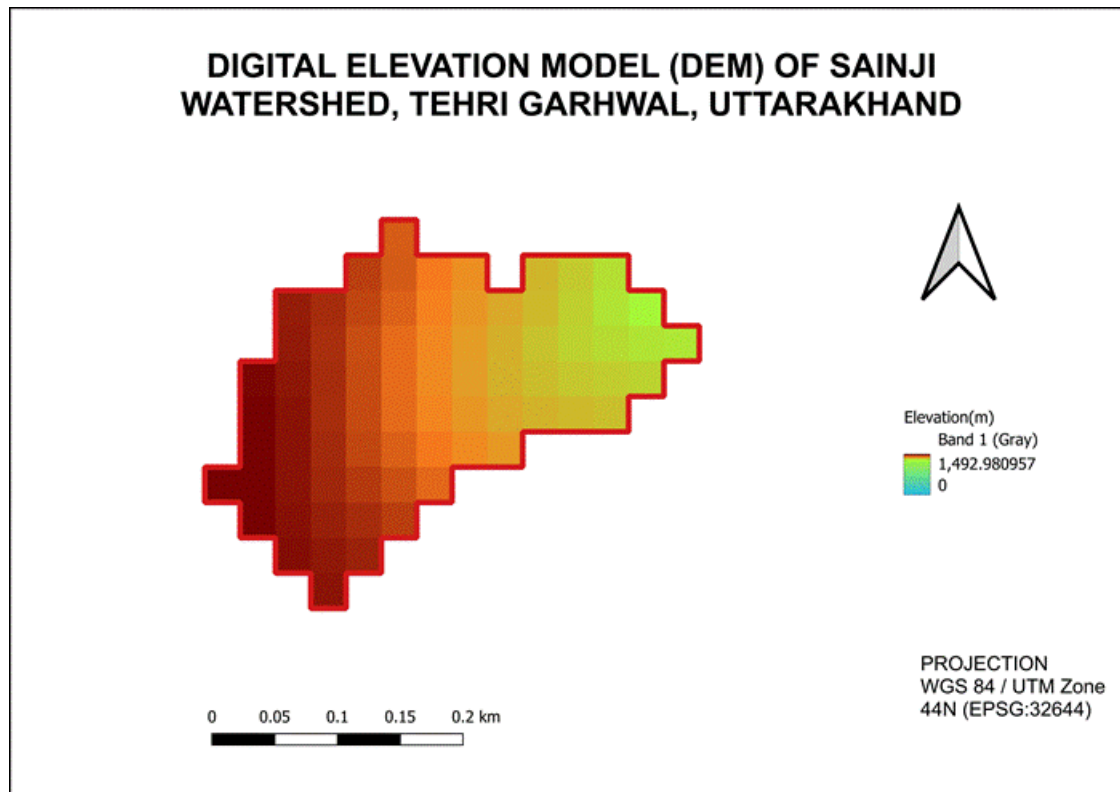


Figure 2: Digital Elevation Model (DEM) of Sainji Watershed, Tehri Garhwal, Uttarakhand

The elevation surface for the watershed ranges from roughly 1,490 m down toward the lower reaches near the outlet, with the highest ground concentrated in the western portion of the catchment (shown in dark red) and the terrain dropping steadily toward the northeast (shown in green). This west-to-east elevation gradient shapes the direction of surface flow across the entire watershed and is the raw input from which every other terrain-derived layer slope, drainage, and stream distance was generated.

5.2 Slope

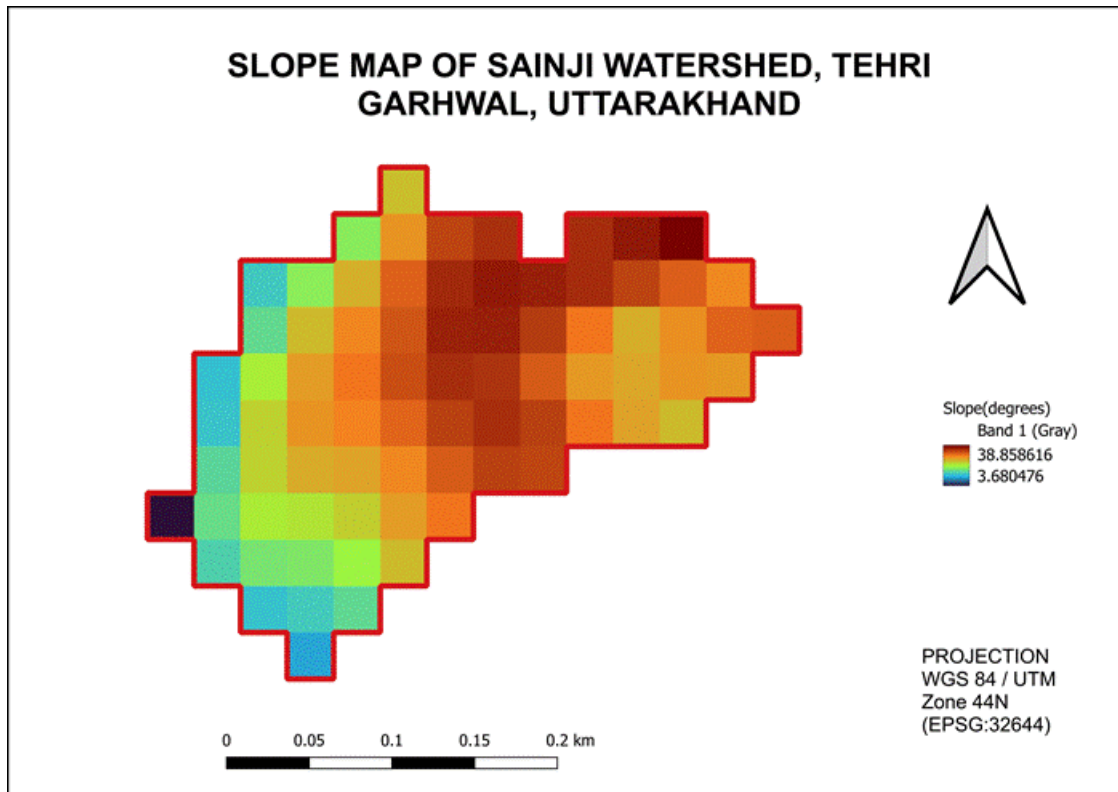


Figure 3: Slope Map of Sainji Watershed, Tehri Garhwal, Uttarakhand

The values for the slope across the whole basin range from 3.7 degrees in the flat regions to as much as 38.9 degrees in the steep regions. As such, the steep slope regions, denoted by dark red and brown colors, are found in the central and upper parts of the drainage basin. The flat slope regions, denoted by blue and teal colors, are mainly located on the west and south sides of the drainage basin. Slope is one of the most important factors (weight of 0.4), meaning that the flat regions will be more suitable.

5.3 Land Use / Land Cover (LULC)

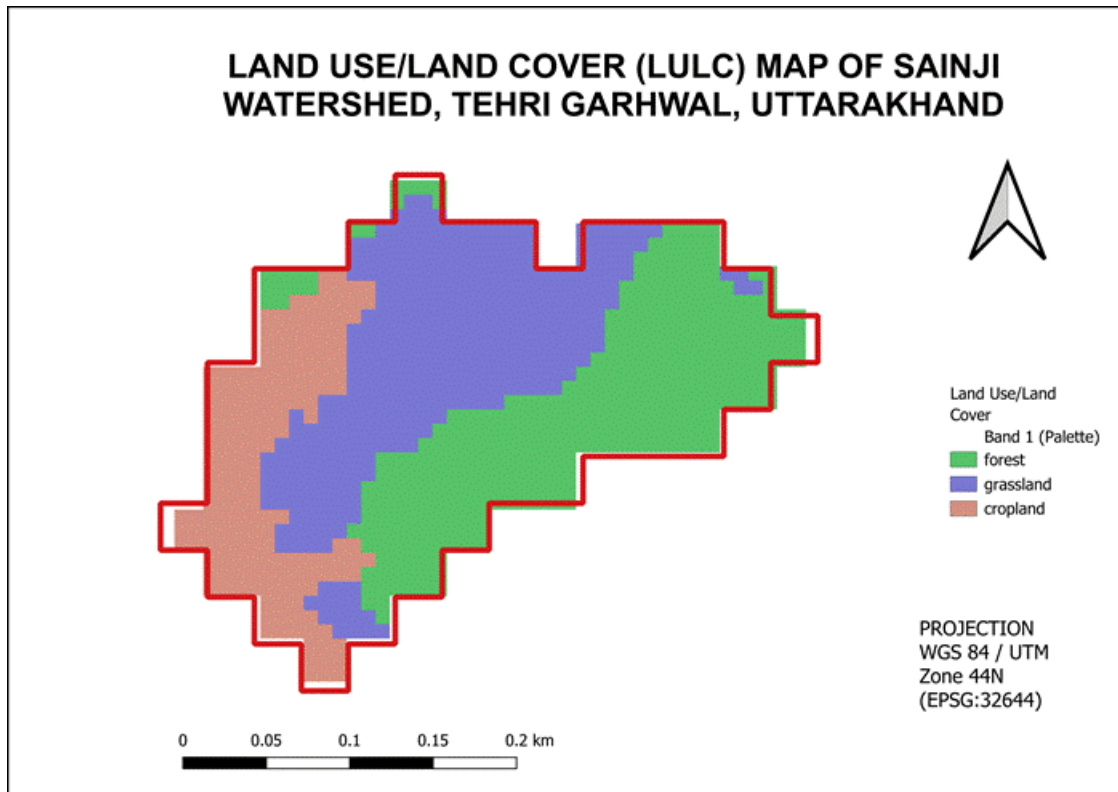


Figure 4: Land Use / Land Cover (LULC) Map of Sainji Watershed, Tehri Garhwal, Uttarakhand

The Sentinel-2-derived LULC classification identifies three dominant classes across the watershed: forest (green), grassland (purple), and cropland (salmon/pink). Cropland is concentrated along the western edge of the watershed, closely matching the terraced fields visible in the satellite base image in Chapter 2. Grassland occupies much of the central and northern portion, while forest dominates the eastern half of the catchment. This spatial pattern matters for recharge because forest and grassland cover generally support better infiltration than intensively cultivated or compacted cropland, though the strength of that relationship also depends on local soil texture, which was outside the scope of the datasets used in this study.

5.4 Stream Distance

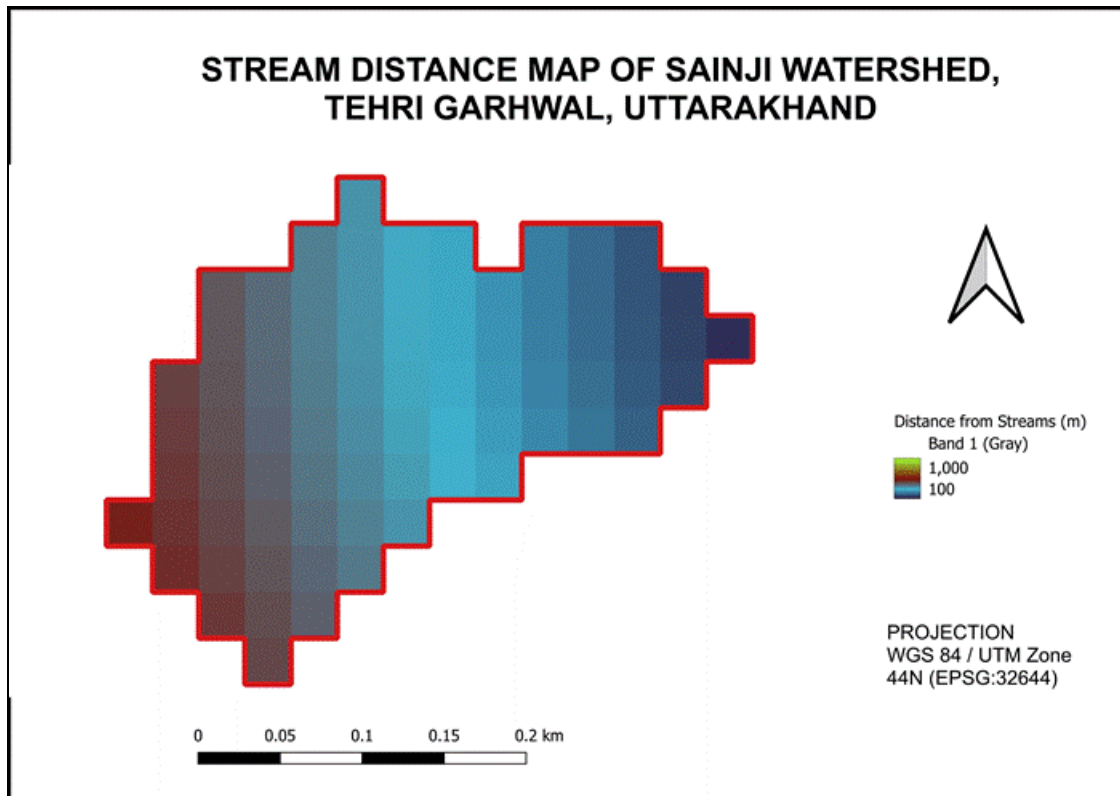


Figure 5: Stream Distance Map of Sainji Watershed, Tehri Garhwal, Uttarakhand

The stream distance layer, created as a Euclidean distance surface from the DEM-extracted drainage network, shows that the western part of the watershed is much farther from the nearest stream channel. This area is marked in dark red and can be up to roughly 1,000 m away. In contrast, the central and eastern zones are within about 100 m of a stream, indicated in blue. Since proximity to drainage is the second most important factor in the overlay, with a weight of 0.3, the central and eastern parts of the watershed are likely to score better on this factor, even where the slope conditions are not ideal.

5.5 Groundwater Recharge Suitability Map (Continuous)

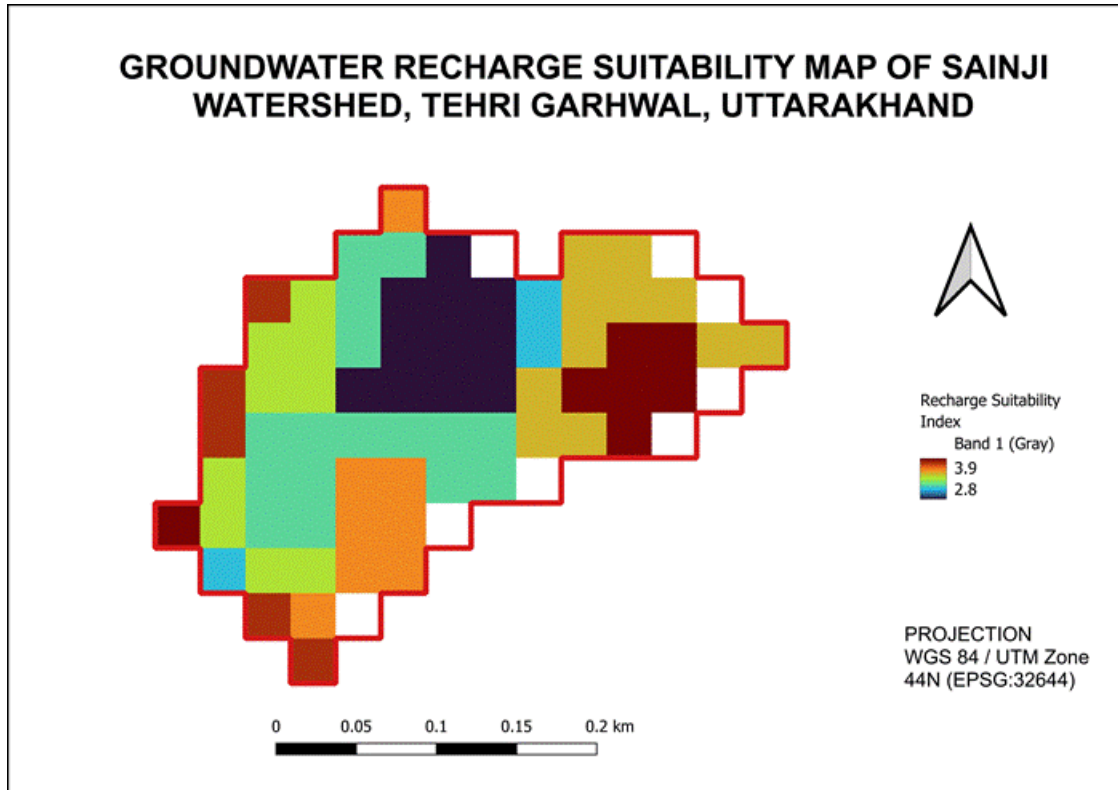


Figure 6: Groundwater Recharge Suitability Map of Sainji Watershed (continuous suitability index, 2.8-3.9)

This is the direct result of the weighted overlay procedure described in Chapter 4. It is a continuous surface with each cell having a suitability index from approximately 2.8 to 3.9. The pattern is clearly patchy rather than smoothly graded, and this directly reflects the underlying terrain: the small watershed such as Sainji has sharp local transitions in slope, land cover and stream proximity, and the weighted overlay reproduces those transitions rather than smoothing them out. The least suitable area is the deep purple zone in the upper-central part of the watershed, which corresponds to the steep slopes shown in Figure 3. The warmer orange and red patches in the east are areas where slope, proximity to the stream and land cover are more favourable.

5.6 Recharge Classification Map

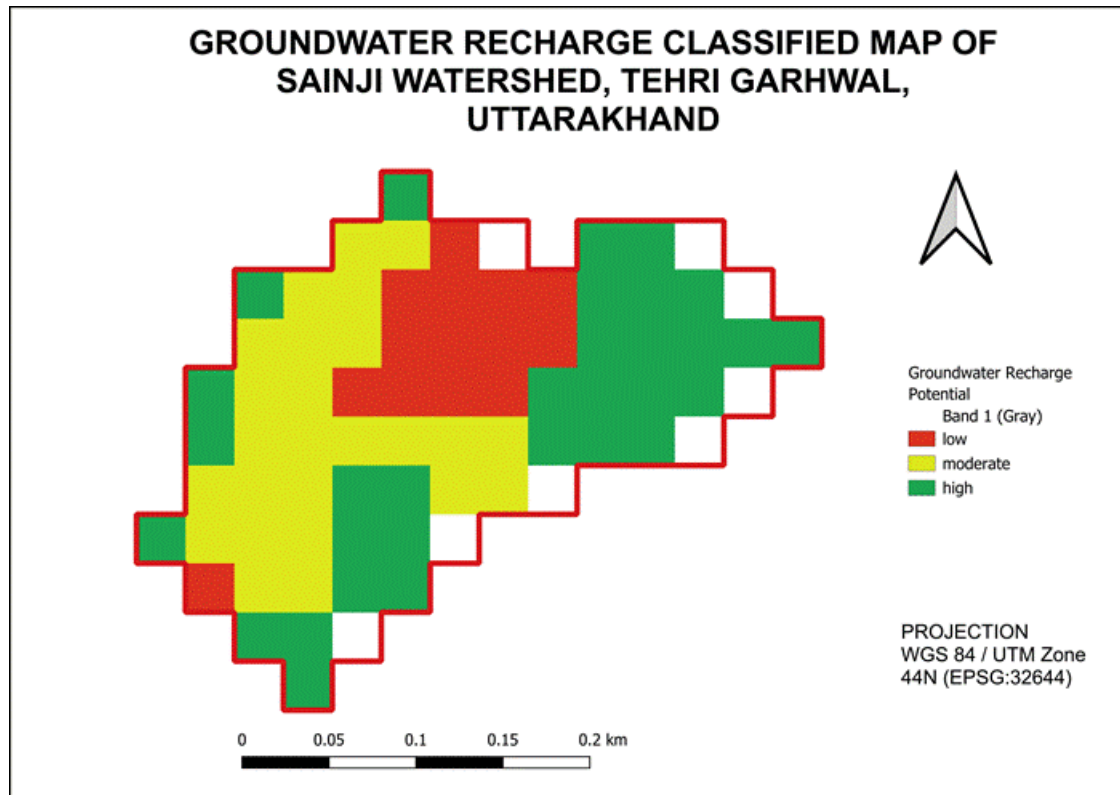


Figure 7: Groundwater Recharge Classified Map of Sainji Watershed (Low / Moderate / High)

To aid in the understanding and application of the suitability surface, the continuous index presented in Figure 6 was reclassified into three suitability classes: low (red), moderate (yellow) and high (green). This is probably the most useful output of the GIS analysis for practical decision making, as this directly answers the question where to locate water harvesting structures such as farm ponds, percolation pits or contour trenches. The high suitability zones (green) are generally along the western and eastern edges of the watershed, and a band of low suitability (red) crosses the upper-central part of the watershed, reflecting the steep terrain shown in the slope map. In between is the moderate class (yellow) a transitional zone which takes a substantial share of the watershed's central area.

Read together, Figures 2 through 7 tell a coherent physical story: the steep, mid-elevation terrain in the centre of the watershed sheds water too quickly for good recharge, while the gentler ground near the drainage network and along the watershed edges much of it under forest or grassland cover offers the best conditions for water to infiltrate and recharge the subsurface. This is precisely the kind of spatially explicit

insight that manual field assessment alone would struggle to produce with the same consistency across an entire watershed.

Chapter 6: Machine Learning Analysis

The GIS-based weighted overlay in Chapter 5 gives a physically-grounded recharge suitability map, but it relies on the weights (0.4 / 0.3 / 0.2 / 0.1) being chosen correctly. Machine learning offers a way to test this from a different angle: rather than assuming a linear weighting scheme, a Random Forest model can learn the relationship between the input factors and recharge suitability directly from the sampled data, and its feature importance output tells us which factors the data itself suggests are most influential.

6.1 Dataset Preparation

The point-sampled data set exported from QGIS (Chapter 4) contained 300 raw records in 12 columns with duplicate feature-ID fields, a legacy of the point-sampling process. The four relevant columns (Stream, Slope, Lulc and Recharge) were selected and incomplete records were removed. 47 rows (15.7%) of the dataset were dropped due to missing Lulc or Recharge values. 253 complete samples were available for modelling. The clean data set was divided into training and testing data with a ratio of 80:20, where there were 202 training samples and 51 testing samples.

Step	Detail
Raw dataset	300 samples × 12 columns
Features used	Stream, Slope, Lulc
Target variable	Recharge
Rows removed (missing data)	47 (15.7%)
Clean dataset	253 samples
Train / Test split	202 / 51 (80:20)

An exploratory look at the cleaned data (Figure 8) shows the distribution of Recharge scores clustering fairly tightly around the 3.2–3.5 range, with fewer samples at the

extremes consistent with the moderate-suitability band dominating the classified map in Chapter 5.

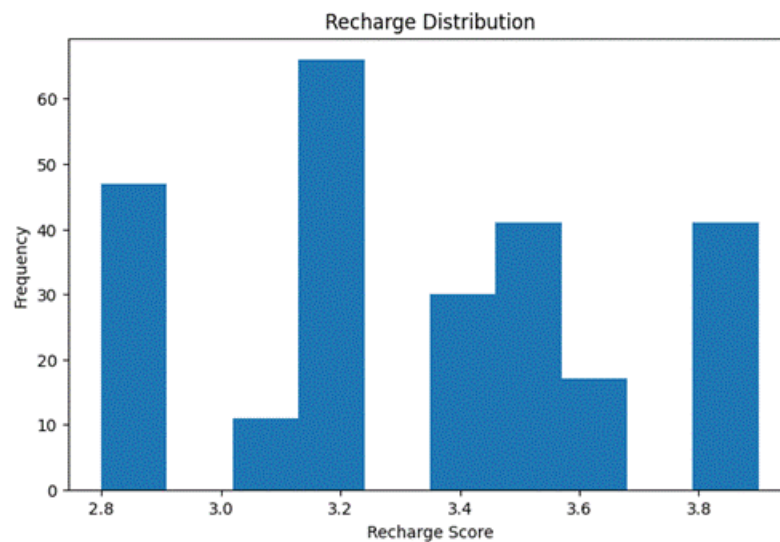


Figure 8: Distribution of Recharge Scores in the Cleaned Training Dataset

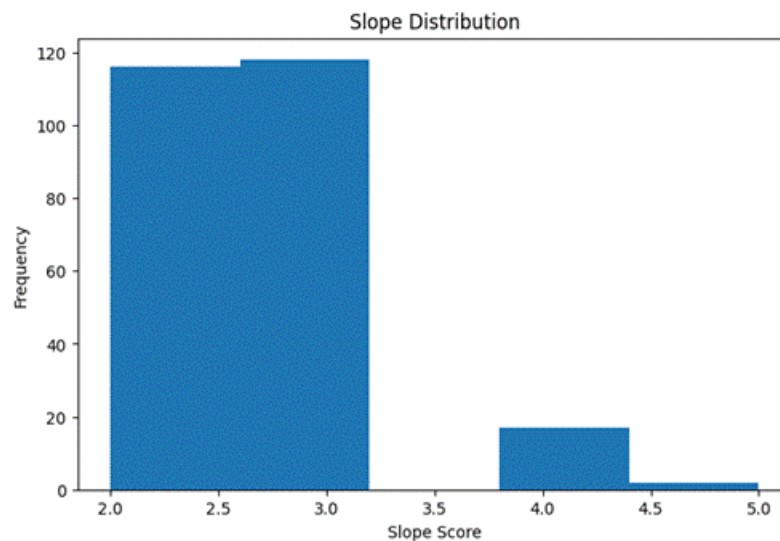


Figure 9: Distribution of Slope Scores in the Cleaned Training Dataset

6.2 Correlation Analysis

A Pearson correlation matrix was calculated across Stream, Slope, Lulc, and Recharge before training the model to determine the strength of the relationship between the three predictor variables and the target as well as the relationship between the predictor variables themselves. Lulc had the highest correlation with Recharge, followed by Slope

and then Stream, a pattern that, as will be shown by the feature importance results below, carries directly through to the trained model.

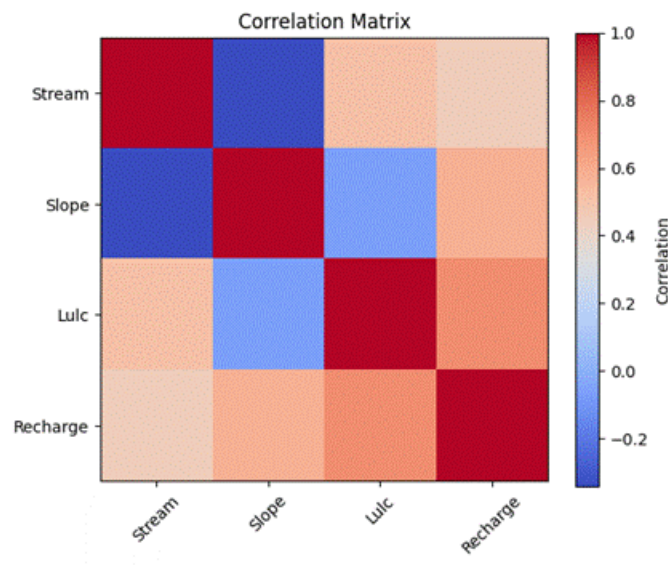


Figure 10: Correlation Matrix Heatmap (Stream, Slope, Lulc, Recharge)

6.3 Random Forest Model — Baseline

A Random Forest Regressor with 100 trees ($n_estimators = 100$, $random_state = 42$) was trained on the 202 training samples using Stream, Slope, and Lulc as predictors of Recharge. On the 51 held-out test samples, the baseline model achieved the following performance:

Metric	Value
Mean Absolute Error (MAE)	0.0025
Root Mean Squared Error (RMSE)	0.0164
R ² Score	0.9975

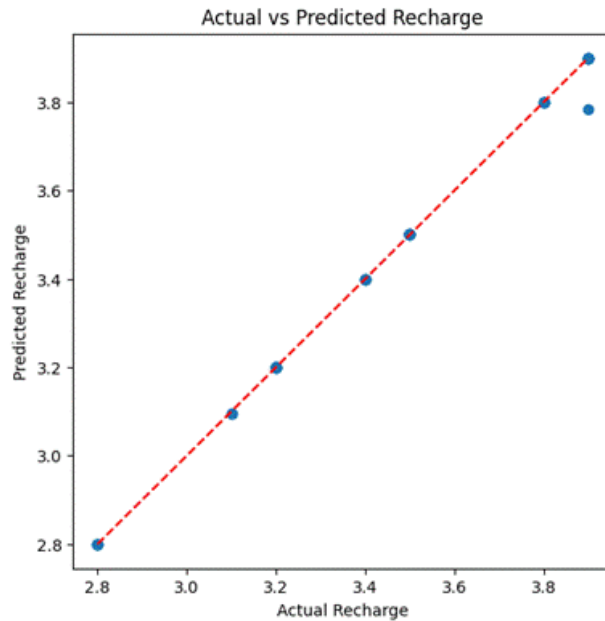


Figure 11: Actual vs Predicted Recharge — Baseline Random Forest Model

The scatter plot in Figure 11 shows predicted values sitting almost exactly on the 1:1 line against actual values, an extremely tight fit for what is, in principle, a fairly noisy real-world prediction problem. This is the first hint of something worth flagging explicitly, which Section 6.6 addresses directly rather than glossing over.

6.4 Feature Importance — Baseline Model

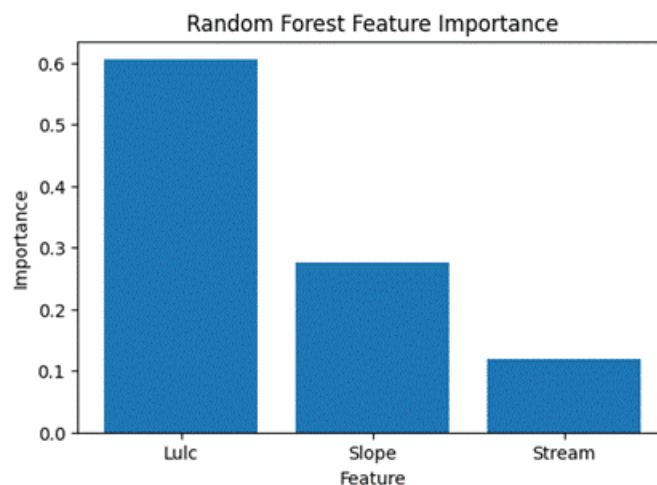


Figure 12: Random Forest Feature Importance — Baseline Model

Lulc ranked first in the feature importance of the baseline model (around 60.5% of total importance), followed by Slope (around 27.7%) and Stream distance (around 11.8%) as

the least important. Interestingly, this ranking is not quite the same as the weights used in the GIS overlay where Slope was given the highest weight (0.4). Such a divergence between a fixed-weight overlay and a data-driven importance ranking is a genuinely useful cross-check , it does not mean either approach is wrong , but it is worth discussing rather than ignoring . Section 6.6 returns to why this divergence should be interpreted cautiously given how the target variable was constructed .

6.5 Hyperparameter Tuning and Cross-Validation

To check whether the baseline model's performance could be improved and to get a more robust estimate of accuracy, two additional steps were carried out. First, 5-fold cross-validation on the baseline model produced R^2 scores of [0.99999, 0.99972, 0.99976, 0.99907, 0.97143], averaging 0.994 consistently high across folds, with only the fifth fold showing a mild dip. Second, a grid search (GridSearchCV, 5-fold cross-validation, scored on R^2) was run over combinations of `n_estimators` (50/100/200), `max_depth` (None/5/10), and `min_samples_split` (2/5). The best configuration found was `n_estimators` = 200, `max_depth` = None, `min_samples_split` = 2, with a best cross-validated R^2 of 0.990.

The tuned model (`best_rf`), refit on the training data with these parameters, produced a modest improvement on the test set:

Metric	Baseline RF	Tuned RF (GridSearchCV)
MAE	0.0025	0.0022
RMSE	0.0164	0.0142
R^2	0.9975	0.9981

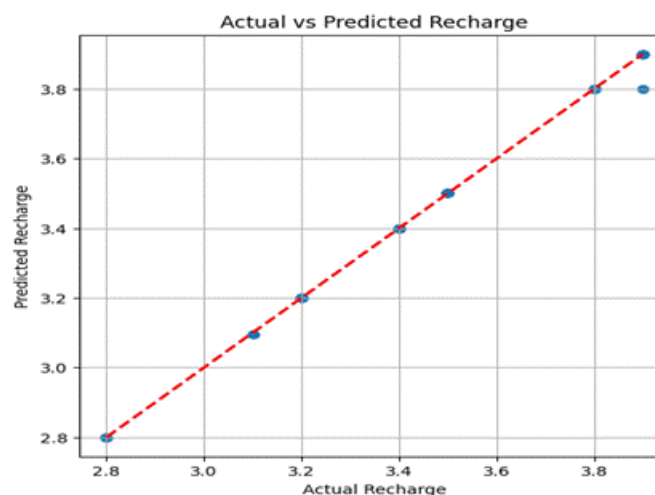


Figure 13: Feature Importance — Tuned Random Forest Model

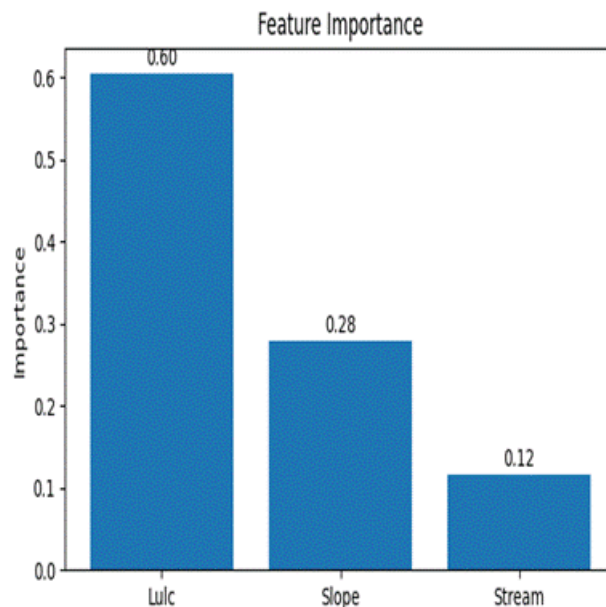


Figure 14: Actual vs Predicted Recharge — Tuned Random Forest Model

The tuned model's feature importance ranking (Lulc 60.4%, Slope 27.9%, Stream 11.7%) is essentially unchanged from the baseline, confirming that the ranking is a stable property of the dataset rather than an artefact of a particular set of hyperparameters. The final tuned model was saved as `Groundwater_RF_Model.pkl`, and its predictions on the test set were exported to `Groundwater_Recharge_Predictions.csv` for reference.

6.6 Interpreting the Results — Why the Accuracy Is (Almost Too) High

An R^2 above 0.99 is, frankly, higher than a machine learning model should normally achieve on real-world environmental data, and it deserves an honest explanation rather than a celebratory one. The reason is straightforward once stated plainly: the Recharge values used as the model's target were themselves calculated in QGIS as a weighted combination of Slope, Stream distance, and LULC (plus rainfall, which was not included as a machine learning feature). In other words, the model was not asked to predict an independently observed quantity, such as a field-measured water table depth or spring discharge; it was asked to recover a formula that had already been used to generate its own training labels from three of its own input features.

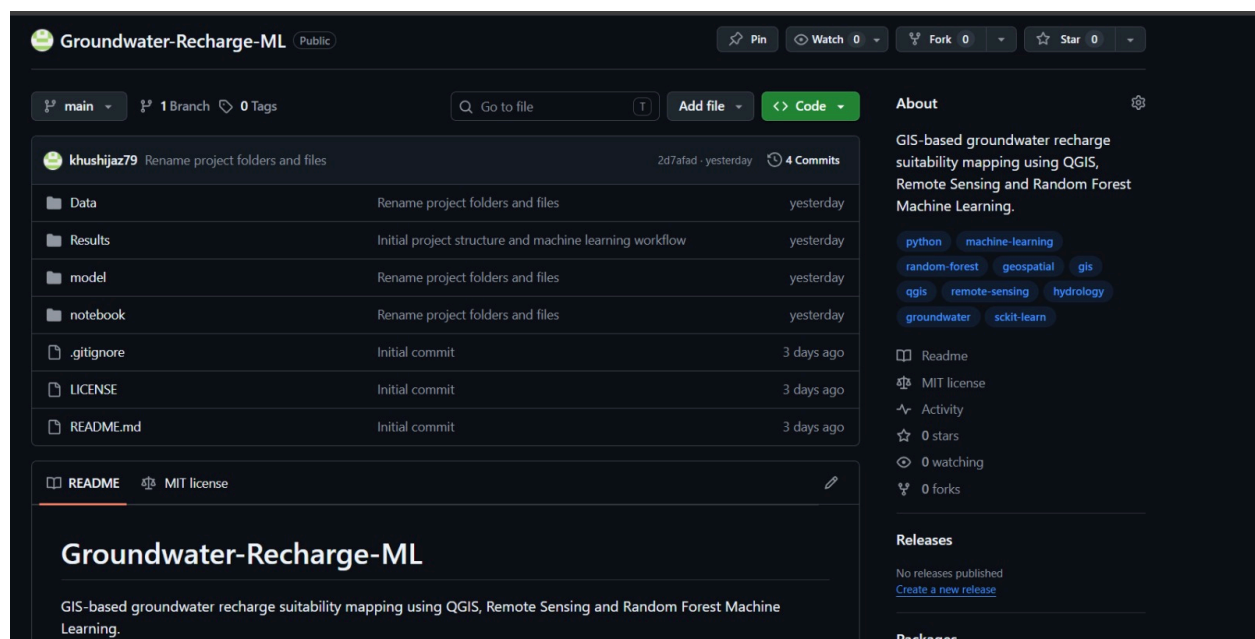
Given that setup, a Random Forest with enough trees and depth is very well suited to reconstructing a smooth, largely linear weighted-sum relationship almost exactly, which is exactly what happened here. This does not make the machine learning exercise pointless: the feature importance ranking still offers a legitimate, data-driven cross-check on which factors dominate the relationship (and here, LULC turns out to matter more than the fixed 0.2 GIS weight might suggest), and the exercise demonstrates a complete, reproducible ML pipeline data cleaning, train/test splitting, cross-validation, hyperparameter tuning, and model persistence. But the headline R^2 of 0.998 should not be read as evidence that the model can predict real, field-observed groundwater recharge with near-perfect accuracy. It is evidence that the model can recover a known formula from its own inputs, which is a materially different and more modest claim. This limitation is discussed further in Chapter 8 and should be stated explicitly in any presentation or viva defence of this project.

Chapter 7: GitHub Repository and Reproducibility

One of the explicit goals of this project was to make the entire workflow GIS processing, machine learning, and this report reproducible by anyone with access to a computer and an internet connection, without the need for licensed software. To achieve this, the project is organised as a version-controlled GitHub repository with the following structure:

```
Groundwater-Recharge-ML/  README.md  requirements.txt  data/
model/  results/  notebook/  workflow/  report/  docs/
```

The README.md documents the project's purpose, datasets, and how to reproduce each step; requirements.txt lists the exact Python package versions needed to rerun the machine learning notebook; data/ holds the raw and cleaned CSV files; model/ holds the saved Random Forest model (Groundwater_RF_Model.pkl); results/ holds the exported prediction CSV and figures; notebook/ holds the Google Colab notebook covering the full ML pipeline described in Chapter 6; workflow/ documents the QGIS processing steps and, where possible, exported QGIS models or processing scripts; and report/ and docs/ hold this written report and any supplementary documentation.



Publishing the project this way means that another student, or the same student revisiting the project a year later, can clone the repository, install the dependencies from

requirements.txt, and rerun the entire analysis from raw data to final maps and model without needing access to any proprietary software or paid data subscriptions everything used here (QGIS, Python, scikit-learn, SRTM, Sentinel-2, NASA POWER) is free and openly licensed.

Chapter 8: Conclusion and Future Scope

8.1 Conclusions

- The Sainji watershed was successfully delineated from SRTM DEM data and its slope, stream network and LULC characteristics were mapped by using totally open source tools and free datasets.
- The groundwater recharge suitability map prepared by four factor weighted overlay (slope 0.4, stream distance 0.3, LULC 0.2, rainfall 0.1) demarcates different high, moderate and low suitability zones in the watershed, where the steep mid-elevation terrain is least conducive area for recharge.
- However, as discussed in Chapter 6, the Random Forest model trained on GIS-sampled data showed very high predictive accuracy ($R^2 \approx 0.998$) because it recovered the known weighted-overlay formula from its own inputs and not because of an independent validation against field-measured groundwater data. This is an important and honest caveat, not a failing of the analysis.
- The machine learning feature importance analysis (LULC $\approx 60\%$, Slope $\approx 28\%$, Stream $\approx 12\%$) offers a useful, data-driven cross-check against the GIS-assigned weights, and suggests that land cover may deserve more emphasis in future iterations of the overlay than the current 0.2 weighting gives it.
- The complete workflow from raw DEM to final suitability map and trained model has been documented and version-controlled on GitHub, making it fully reproducible and adaptable to other small Himalayan watersheds facing similar water security challenges.

8.2 Future Scope

- Incorporate soil texture and geology datasets that strongly influence infiltration but were not available for this study to improve both the GIS overlay and the machine learning feature set.
- Alternatively, instead of assuming weights and using a fixed-weight overlay, a more formal pairwise comparison using the Analytic Hierarchy Process (AHP) approach (Zimik et al., 2022, as performed for the Sikkim Himalaya recharge study) can be utilized to derive the weights.
- Conduct field validation of the recharge suitability map using actual data of discharge from springs, water table level in wells, or piezometer readings that would enable training of the machine learning algorithm on independent target variables, solving the primary issue highlighted in Chapter 6.
- Compare Random Forest against other algorithms such as Gradient Boosting or XGBoost, which have shown marginally stronger performance in comparable rainwater harvesting suitability studies when properly validated against independent field data (Asrade et al., 2026).
- Extend the methodology to the larger watershed system surrounding Sainji, to assess whether the same weighting scheme and factor relationships hold at a broader scale.

References

Asrade, T. M., Abebe, S. A., & Tadesse, K. B. (2026). Rainwater harvesting potential zone identification using GIS – Machine learning approach and comparison of their performance. *Agricultural Water Management*, 324, 110134. <https://doi.org/10.1016/j.agwat.2026.110134>

Zimik, H. V., Angchuk, T., Misra, A. K., Ranjan, R. K., Wanjari, N., & Basnett, S. (2022). GIS-based identification of potential watershed recharge zones using analytic hierarchy process in Sikkim Himalayan region. *Applied Water Science*, 12, 248. <https://doi.org/10.1007/s13201-022-01758-5>

NASA Jet Propulsion Laboratory. Shuttle Radar Topography Mission (SRTM) Digital Elevation Data. <https://www2.jpl.nasa.gov/srtm/>

European Space Agency. Sentinel-2 Mission Copernicus Open Access Hub. <https://sentinel.esa.int/web/sentinel/missions/sentinel-2>

NASA POWER Project. Prediction of Worldwide Energy Resources Precipitation Data. <https://power.larc.nasa.gov/>

QGIS Development Team. QGIS Geographic Information System. Open Source Geospatial Foundation Project. <https://qgis.org>

Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Appendix

A.1 Machine Learning Pipeline — Code Overview

The full Google Colab notebook (Groundwater.ipynb) is included in the project's GitHub repository under notebook/. Its structure, summarised here for reference, follows the standard flow described in Chapter 6:

- Data loading and inspection (pandas: head, shape, columns, info, describe, isnull, duplicated)
- Column selection and cleaning (Stream, Slope, Lulc, Recharge; dropna)
- Exploratory data analysis (histograms of Recharge and Slope distributions)
- Correlation matrix and heatmap
- Feature/target split (X = Stream, Slope, Lulc; y = Recharge) and 80:20 train-test split
- Random Forest training ($n_estimators = 100$, $random_state = 42$)
- Evaluation (MAE, RMSE, R^2) and actual-vs-predicted scatter plot
- Feature importance extraction and bar chart
- 5-fold cross-validation
- Hyperparameter tuning via GridSearchCV ($n_estimators$, max_depth , $min_samples_split$)
- Refit of the best model, final evaluation, and updated feature importance / scatter plot
- Model persistence (joblib.dump) and export of final predictions to CSV